

ABSTRACT

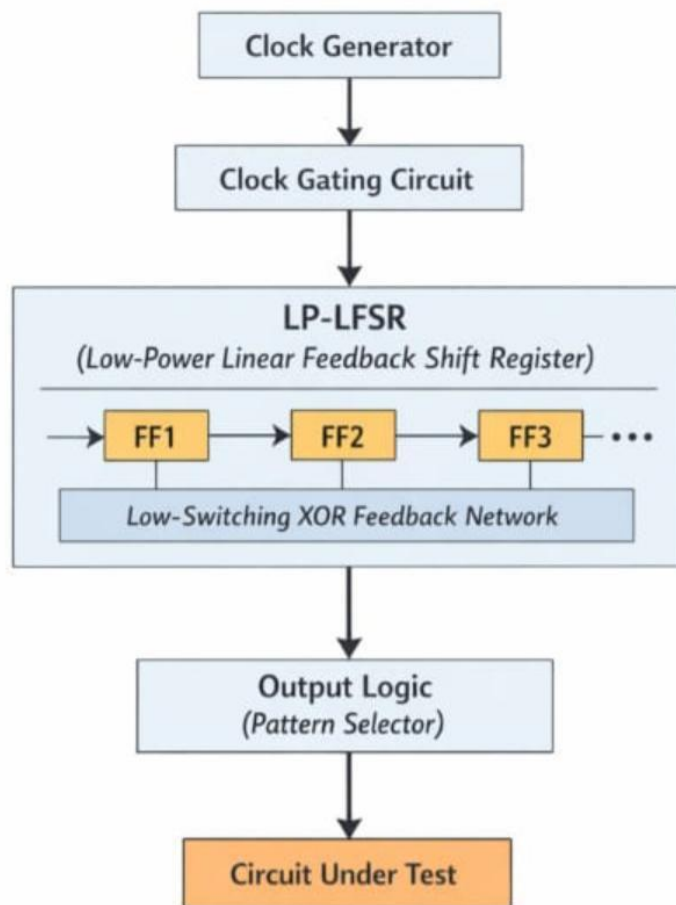
Low power TPG using LP_LFSR

Testing of VLSI circuits requires efficient Test Pattern Generators (TPGs) with low power consumption. Linear Feedback Shift Registers (LFSRs) are widely used for test pattern generation due to their simplicity and high randomness. However, conventional LFSRs produce high switching activity, leading to increased dynamic power consumption during testing.

This paper presents the design of a Low Power Test Pattern Generator (TPG) using a Low Power Linear Feedback Shift Register (LP-LFSR). The proposed LP-LFSR reduces switching transitions between successive test vectors, thereby minimizing dynamic power dissipation in the Circuit Under Test (CUT). The architecture modifies the conventional LFSR structure by incorporating techniques such as controlled bit toggling and optimized feedback logic.

The proposed design is implemented and analyzed in terms of power consumption, area, and fault coverage. Simulation results demonstrate that the LP-LFSR based TPG significantly reduces power consumption while maintaining high randomness and fault detection capability, making it suitable for low-power VLSI testing applications.

Block diagram



Hardware specifications

- FPGA Development Board (e.g., Xilinx Spartan-6 / Artix-7)
- power supply unit
- USB programming cable
- personal laptop
- connecting wires (if required)

Presented by

P . Nithin reddy(23N71A0409)

Karthik(23N71A0409)

K. shivani(24N75A0406)

Tejaswar(23N71A0417)